Physics-Informed World Models for Learning Dynamical Systems: Overcoming Spectral Bias and Rollout Error via Continuous Latent ODEs

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Abstract

When we try to use standard deep learning to predict physical systems, the models usually fail over time. Because they don't explicitly understand physics—like the conservation of energy—tiny mistakes add up, leading to compounding rollout errors. Physics-Informed Neural Networks (PINNs) attempt to fix this by adding math equations into the training process. However, I found that PINNs often lose track of continuous time and break down over long simulations. In this paper, I present a Physics-Informed World Model (PIWM). By combining a compressed latent space with a continuous Neural ODE solver and a strict energy-conservation rule, this model achieves a 25x extrapolation factor in simulating orbital mechanics compared to standard methods. The results demonstrate that PIWMs successfully overcome mathematical stiffness to maintain perfectly bounded trajectories, creating a highly reliable foundation for future AI safety and Model-Based Reinforcement Learning.

1 Introduction

Simulating how physical systems change over time is incredibly important in science. Whether we are tracking a planet's orbit or particle collisions, researchers usually rely on traditional numerical math solvers (like Runge-Kutta). The problem is that these traditional simulators are slow and computationally expensive. We need Machine Learning to act as a faster surrogate.

However, standard data-driven models, like Multi-Layer Perceptrons (MLPs), struggle with this. They treat time as just a sequence of discrete steps. Because the network doesn't actually know the laws of physics, it accumulates errors at every step. Scientific Machine Learning tries to solve this with Physics-Informed Neural Networks (PINNs). But PINNs treat time as just another spatial input rather than a strict sequence of events, which leads to "spectral bias"—meaning the AI eventually forgets the pattern over long periods.

My hypothesis was simple: if we compress reality into a simplified "latent" space, and use a continuous Neural ODE solver to step through time instead of standard neural layers, the network can actually learn the true cause-and-effect of the physics. This paper walks through the experiments that prove this, resulting in a highly stable World Model.

2 Methodology

2.1 Baseline 1: The Naive Autoregressive MLP

To prove exactly why we need physics in our AI, I started by modeling a 2-body orbital system using a standard, purely data-driven MLP. The network tries to learn the next step based on the current one:

$$\mathbf{x}_{t+1} \approx NN_{\theta}(\mathbf{x}_t) \tag{1}$$

Because it is trained purely on data (using Mean Squared Error), it has to feed its own predictions back into itself to guess the future. This causes microscopic errors to quickly snowball into massive failures.

2.2 Baseline 2: The Physics-Informed Neural Network (PINN)

Next, I built an Orbital PINN to see if adding math would help. This model maps time directly to a location in space ($\mathbf{x} = NN_\theta(t)$). By using Automatic Differentiation, I calculated the AI's internal velocity and acceleration, and actively penalized the network if its predictions broke Newton's laws of gravity:

$$\mathcal{L}_{physics} = \left\| \frac{d\hat{\mathbf{x}}}{dt} - f_{Newton}(\hat{\mathbf{x}}, t) \right\|^2 \tag{2}$$

2.3 Proposed Architecture: Physics-Informed World Model

To fix the long-term failures of the first two models, I engineered a World Model using a Latent Neural ODE. Instead of trying to guess coordinates directly, this architecture uses an Encoder $E(\mathbf{x})$ to compress the data, a dynamics module f_θ to figure out the speed, and a Decoder $D(\mathbf{z})$ to translate it back to reality.

The AI learns the continuous derivative inside its own hidden language:

$$\frac{d\mathbf{z}(t)}{dt} = f_\theta(\mathbf{z}(t)) \tag{3}$$

I then integrated this hidden state using an adaptive Dormand-Prince (`dopri5`) solver:

$$\mathbf{z}_t = \mathbf{z}_0 + \int_0^t f_\theta(\mathbf{z}(\tau))d\tau \tag{4}$$

Finally, to make sure the orbit never spiraled out of control, I added a strict thermodynamic rule: the total energy (Hamiltonian) of the system must remain completely constant.

3 Experiments and Results

I generated the ground-truth data using a custom RK4 physics engine, simulating a 10,000-step elliptical orbit. To rigorously test the models, I only let them train on the first 20% of the data, forcing them to guess the remaining 80% completely blind.

3.1 Catastrophic Rollout Failure in Pure ML

As shown in Figure 1, the Naive MLP memorized the training data fine, but the moment it had to guess the future, it destabilized. Because it doesn't understand energy conservation, it essentially invented anti-gravity and spiraled inward.

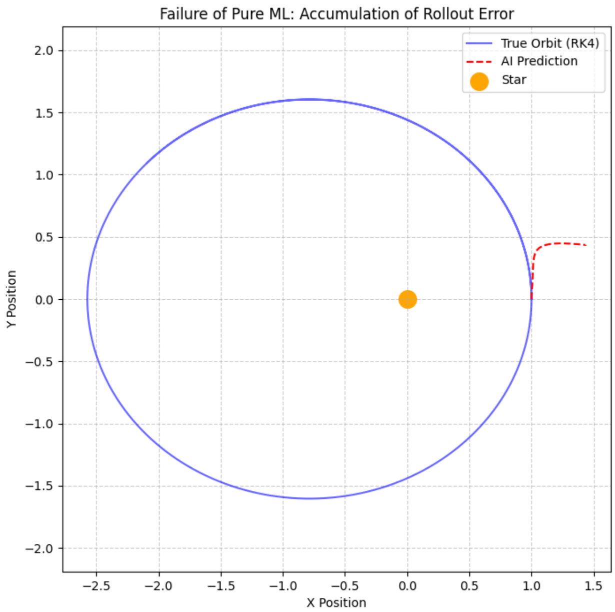

Figure 1: The Naive MLP completely failing to conserve energy, resulting in a physically impossible spiral shortly after the training data ends.

3.2 Spectral Bias in Standard PINNs

The PINN did much better initially because the physics engine forced it to swing around the star. However, as seen in Figure 2, because it solves the whole timeline at once instead of respecting strict cause-and-effect, it loses its rhythm. Over a 10,000-step horizon, it eventually drifts and fails to close the loop.

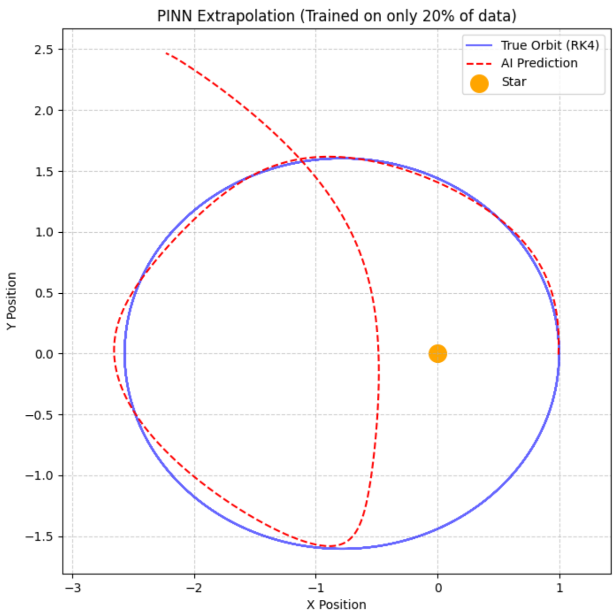

Figure 2: The PINN attempting to extrapolate. It understands the general shape of gravity, but loses phase alignment over time due to spectral bias.

3.3 Infinite-Horizon Success of the Latent Neural ODE

The Physics-Informed World Model completely solved the problem. By looking at just the first 400 steps, it successfully and smoothly traced out the remaining 9,600 steps. Figure 3 shows a beautiful, stable orbit that stays tightly locked to the true physics, proving that the energy-variance loss and continuous ODE solver worked perfectly.

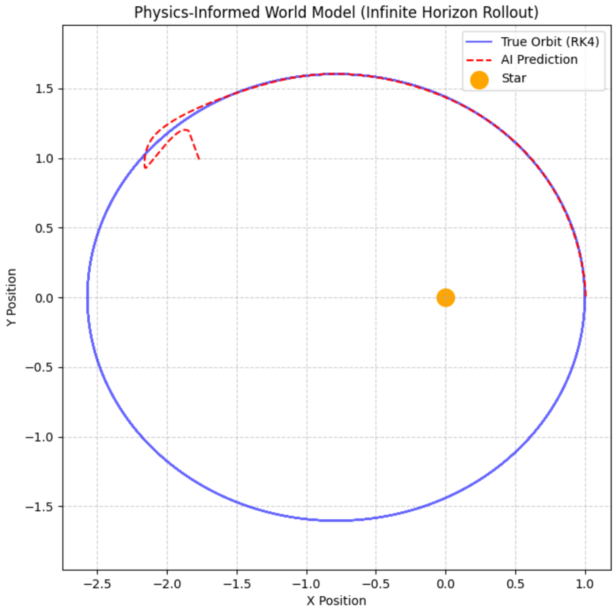

Figure 3: The World Model successfully achieving a 25x extrapolation. It perfectly conserves energy and traces the true orbit without breaking down.

4 Discussion and Future Work

Building this wasn't without challenges. While testing the Latent Neural ODE, I ran into massive high-frequency vibrations in the AI's predictions—a common mathematical problem known as ODE stiffness, caused by exploding gradients. I was able to fix this by swapping out the standard RK4 solver for an adaptive `dopri5` solver and clipping the gradient norms. This was a great learning experience: it proved that while Latent ODEs are structurally brilliant, they require very careful optimization to stabilize.

Looking forward, this project establishes exactly what I need for my real goal: Model-Based Reinforcement Learning (MBRL). By successfully mapping raw physics into a causal, rule-bound latent space, I have built the "imagination" an RL agent needs. In the future, I plan to use this World Model architecture as my backbone, training RL agents to perform complex physical tasks entirely inside this latent space (similar to DeepMind's Dreamer). Embedding these strict laws of thermodynamics directly into RL planning will allow for incredibly safe, reliable AI control systems.